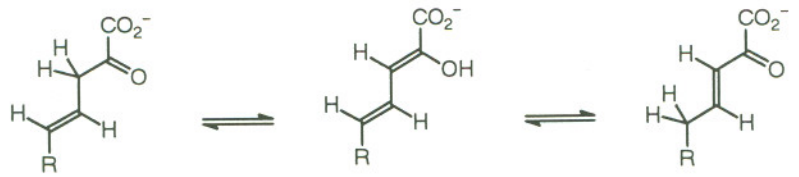
Organic Chemistry Cumulative Examination
October 2, 1993

1. Lian, H; Whitman, C. P. *J. Am. Chem. Soc.* **1993**, *115*, 7978.
This paper describes the study of the following reactions, catalyzed by the enzyme 4-oxalocrotonate tautomerase (4-OT). In the reactions studied, the reactants are the ones in the middle (**1** and **4**), and the possible products are to the left and right.

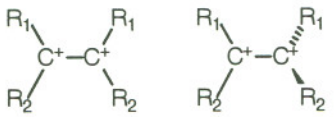


3 : R = CO_2^- **1** : R = CO_2^- **2** : R = CO_2^-
6 : R = H **4** : R = H **5** : R = H

It turned out that the product from **1** was predominantly **2**, while the product from **4** was predominantly **6**. When the reaction was carried out in D_2O, the product from **1** was (5S)-[5-D]-**2**, and the product from **4** was (3R)-[3-D]-**6**.

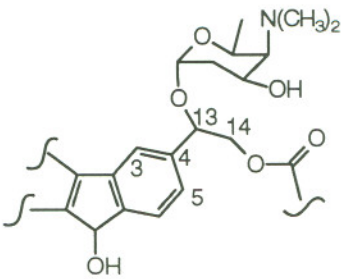
a. Which of the alternative products, **2** vs. **3** and **5** vs.**6**, would you expect to be the more thermodynamically stable? Why?
b. Draw the structures of the deuteriated products, showing the stereochemistry.
c. The physiological function of the enzyme is actually the isomerization from **3** to **2**, effectively a 1,3 hydrogen shift. What is the stereochemistry of the reaction, as demonstrated by these studies?

2. Olah, G. A.; Bausch, J.; Rasul, G.; George, H.; Surya Prakash, G. K. *J. Am. Chem. Soc.* **1993**, *115*, 8060.
Ethylene is planar, obviously because of the existence of the π-bond. In ethylene dication, the two electrons making up the π-bond is removed, and the structure ends up being perpendicular, presumably because of steric interactions between the R groups.



However, calculations described in this paper indicated that diprotonated oxalic acid (R_1=R_2=OH) and diprotonated oxamide (R_1=NH_2; R_2=OH) prefer the planar structure once again.
a. Explain this observation using resonance structures.
b. Explain this observation by drawing the orbital interaction diagram between the two $R_1CR_2^+$ fragments.

3. Leet, J. E.; et al. *J. Am. Chem. Soc.* **1993**, *115*, 8432.
The following is a fragment of a derivative of the kedarcidin chromophore, prepared to elucidate its structure.



NOE's were observed between one of the C-14 protons (call it H-14a) and the C-5 proton, and between the C-13 proton and the C-3 proton. Also, a large coupling (J=3.5Hz) was obseverd between H-14a and the H-13 proton. Determine the stereochemistry at C-13 (R or S?).
Warning: it is possible that I have deliberately drawn the structure above to be mislead you. Don't rely on it!